

Vihan Lalan

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EDUCATION

SVKM's Dwarkadas J. Sanghvi College of Engineering

B.Tech in Computer Engineering (ICB) with Honours in Data Analytics

Mumbai, MH

Aug 2024 – Present

- CGPA - 9.8 (Currently in Sem IV)

WorldQuant University

Intro to Data Science - Foundational Data Science

Remote

Sep 2025 – Present

Friends Academy

High School Diploma

Mumbai, MH

May 2022

EXPERIENCE

Research Consultant

WorldQuant LLC

July 2025 – Present

Remote

- Building and optimizing quantitative ML-driven signals using RSI, MACD, and hybrid indicators, improving Sharpe and fitness ratios by 5–7%.
- Developing financial market prediction strategies combining statistical methods with behavioural market models.

RESEARCH

Operationalizing the Adaptive Market Hypothesis

2026 – Ongoing

- Focused on modeling and operationalizing the Adaptive Market Hypothesis using behavioural and quantitative market indicators.

Quantum Walks on S&P 500 Market Structures

2026 – Ongoing

- Researching the application of quantum walk frameworks on S&P 500 market networks to analyse crash propagation and market instability.

PROJECTS

Directional Prediction Classifier | *Python, scikit-learn, Pandas, NumPy*

May 2026

- Built a machine learning based directional classifier to predict short-term market movement using technical indicators, feature engineering, and classification models with backtesting and evaluation pipelines.

Dynamic Portfolio Rebalancing | *Python, Pandas, NumPy, Matplotlib*

Oct 2025

- Built a dynamic portfolio rebalancing engine optimizing NIFTY 500 allocations using momentum, inverse volatility weighting, and CPPI-based risk management with realistic backtesting constraints.
- Implemented portfolio analytics and allocation logic for adaptive risk-managed investing strategies.

Stock Market Prediction Tool | *Python, yFinance, scikit-learn, Matplotlib*

July 2025

- Developed a stock market prediction system using rolling linear regression models, automated backtesting, RMSE/R² evaluation, and real-time visualization on Yahoo Finance market data.

CERTIFICATIONS

OpenCV: PyTorch Bootcamp

Jan 2026

OpenCV: TensorFlow Bootcamp

Dec 2025

WorldQuant LLC: Gold Level in WorldQuant Challenge

Jul 2025

HackerRank: Programming in Python, Java, and Javascript

Sep 2024

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript

Machine Learning: scikit-learn, PyTorch, TensorFlow, Keras, LSTM/Transformers

Finance & Quant: Backtesting, Quantitative Signals, Portfolio Optimization, Statistical Modelling, Financial Forecasting

Data Tools: Pandas, NumPy, Matplotlib, SQL, yFinance

Web/Software: React.js, Tailwind CSS, Node.js